

Stanislav Perfilyev

Senior C++ Developer · Embedded Systems · Low-Latency · Qt/QML · Python Backend

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PROFESSIONAL SUMMARY

Senior C++/Qt/Python developer with 10+ years of commercial experience in embedded and high-performance software. Built a FreeRTOS-compatible RTOS kernel from scratch (ucontext_t cooperative + priority scheduling, DES time advancement, Mutex/Semaphore/Queue/EventGroup) and a full ARM Cortex-M0 simulator with UART/SPI/I2C/CAN peripherals (all 19 Thumb-16 instruction formats, 165 GTests). Systems Linux developer: eBPF kernel tracer (kprobe/XDP), 4 loadable kernel modules (chardev/procfs/netfilter/kthread). Low-latency specialist: HFT Order Book + Matching Engine with RDTSC timer, std::pmr memory pool, p99 < 200 ns. C++20 expert: Concepts, Ranges, Coroutines (co_yield/co_await), lock-free SPSC queue, memory_order semantics. Robotics: ROS2 robot with computer vision (OpenCV), speech recognition, and LLM-driven dialogue. Quality & infrastructure: custom CI quality-gate system (166+ automated checks — ASAN/TSAN, PVS-Studio, cppcheck, MT-pattern detection, git pre-commit hooks) across 20+ repositories; 450+ automated tests (GTest) across the C++ portfolio. Production backend: FastAPI + PostgreSQL + Redis + Docker on Railway (web-production-f7596.up.railway.app), 100+ pytest tests, 4 GitHub Actions cron bots. English C1.

TECHNICAL SKILLS

C++ Core	C++11/14/17/20/23 · STL · templates · RAII · [[nodiscard]] · noexcept · custom exceptions
C++20	Concepts · Ranges/Views pipelines · Coroutines (co_yield/co_await) · std::jthread · stop_token · latch · barrier
Lock-free/Atom.	SPSC queue · CAS loop · std::atomic · memory_order (relaxed/acquire/release/seq_cst) · ABA · false sharing
Low-latency	RDTSC ns timer · 64B cache-line alignment · std::pmr memory pool · zero-copy · no malloc on hot path
RTOS/Embedded	ARM Cortex-M0 · Thumb-16 ISA · ucontext_t · TCB · Mutex/Semaphore/Queue/EventGroup · FreeRTOS API
Qt / GUI	Qt 5/6 · Widgets · QML/Qt Quick · QThread · QSqlTableModel · QtCharts · QSystemTrayIcon · Q_LOGGING_CATEGORY
Systems/Linux	WinAPI · DLL injection · SetWindowsHookEx · D-Bus (sd-bus) · /proc · POSIX API · MIL-STD-1553B · eBPF/libbpf/XDP · Kernel Modules (ioctl/Netfilter/kthread)
Python Backend	FastAPI · asyncio · asyncpg · SQLAlchemy · Pydantic v2 · Redis · Websocket · APScheduler · Telegram Bot API · pytest
Databases	PostgreSQL · SQLite · Redis · Google Sheets API v4 · asyncpg · SQLAlchemy ORM
DevOps / CI	Docker · docker-compose · Railway (production) · GitHub Actions · CMake 3.20+ · Makefile
Testing	Google Test · Google Benchmark (p50/p95/p99) · DUnit (Pascal) · pytest · mock/monkeypatch · PVS-Studio · ASAN/TSAN · cppcheck
Tools	CLion · Qt Creator · GDB · Git (SSH) · Bash · PowerShell · Protobuf/gRPC · ROS2
Languages	Russian (native) · English C1 (written/technical)

PORTFOLIO — C++ PROJECTS

Low-latency HFT Order Book & Matching Engine

July 2026

Stack: C++20 · std::atomic · memory_order · RDTSC · std::pmr · Google Test · Google Benchmark

github.com/stanislaw-perfilyev/hft-engine

- **Lock-free SPSC ring buffer:** CAS loop, memory_order_acquire/release, 64-byte cache-line padding — zero false sharing.
- **Order Book:** std::map<Price, std::deque<Order>> price levels, O(log N) insert/cancel, O(1) top-of-book.
- **Matching Engine:** FIFO price-time priority, limit / market / cancel orders, partial fill support.
- **Memory Pool:** std::pmr::monotonic_buffer_resource — no malloc on hot path, 1M pre-allocated Order objects.
- **Latency:** RDTSC nanosecond timer · Google Benchmark: p50 = 85 ns, p99 = 195 ns for top-of-book match.
- **50 GTests + benchmarks · CI:** GitHub Actions (GCC-13 / Clang-16 matrix).

ARM Cortex-M0 Simulator + FreeRTOS-compatible RTOS Kernel

July 2026

Stack: C++20 · ARM Thumb-16 ISA · ucontext_t · Intel HEX · Google Test (68 tests)

github.com/stanislaw-perfilyev/mcu-simulator

- Full ARM Cortex-M0 simulator: all 19 Thumb-16 instruction formats, 16 registers, APSR flags (N/Z/C/V), 64-bit cycle counter.
- Complete ISA: ALU (AND/EOR/LSL/LSR/ASR/ADC/SBC/ROR/TST/NEG/MUL/BIC/MVN), LDR/STR/PUSH/POP/LDM/STM, 14 branch conditions, B/BL/BX.
- IMemoryBus abstraction: 32 KB Flash + 16 KB SRAM + bus fault on out-of-bounds; Intel HEX loader with checksum.
- Verification: factorial demo — ARM Thumb assembly in simulator, r1 = 0x78 = 5! in 19 cycles.
- FreeRTOS-compatible RTOS kernel: ucontext_t context switching, Discrete Event Simulation for instant time advancement when all tasks blocked.
- 16 Task Control Blocks (name, priority, state, resume_tick, stack[8192]); states: READY / RUNNING / BLOCKED / SUSPENDED / DELETED.
- Sync primitives: Mutex, Counting Semaphore, typed Queue<T,N> (lock-free SPSC), EventGroup (bit-flag events).
- FreeRTOS-compatible API: xTaskCreate, vTaskDelay, xSemaphoreTake/Give, xQueueSend/Receive, xEventGroupSetBits/WaitBits/ClearBits.
- Demo (blink.cpp): 3 tasks (producer / consumer / watchdog) exchanging via Queue, liveness via EventGroup.
- Memory-mapped peripherals: UART (TX/RX ring buffers), SPI + W25Q32 Flash emulation, I2C + LM75 temperature sensor emulation, CAN 2.0A; sensor_demo.cpp polls LM75 over I2C in real time.
- 165 GTests (CPU core, memory subsystem, Intel HEX loader, RTOS scheduler, sync primitives, UART/SPI/I2C/CAN peripherals) — all passing.

C++20/23 Showcase — Concepts · Ranges · Coroutines · Lock-free · Atomic

July 2026

Stack: C++20/23 · GTest · Google Benchmark · GitHub Actions (GCC-13 / Clang-16 matrix)

github.com/stanislaw-perfilyev/cpp20-showcase

- **Concepts:** TypedContainer with Numeric + Printable concepts; compile-time constraints via static_assert.
- **Ranges:** lazy views pipeline (filter | transform | take), zero-copy collection processing.
- **Coroutines:** Generator<T> (co_yield), Task<T> (co_await) — async generators without Boost.
- **jthread:** cooperative cancellation (stop_token + stop_callback), std::latch, std::barrier.
- **Lock-free:** SPSC queue on std::atomic + CAS; capacity = power of 2 (bitmask vs modulo).
- **Atomic model:** acquire/release/seq_cst examples, ABA problem, spurious wakeup.

Qt6 Job Search Dashboard — QtCharts · QSqlTableModel · QProcess

June–July 2026

Stack: Qt 6.5 · C++17 · QtCharts · PostgreSQL · QSqlTableModel · GitHub Actions CI

github.com/stanislaw-perfilyev/job-search-automation-portfolio/tree/main/portfolio/qt-job-dashboard

- **QMainWindow + QStackedWidget:** vacancy table (QTableView + QSqlTableModel + QPSQL), color-coded rows, filters.
- **QtCharts:** QLineSeries (responses/day), QPieSeries (source split), QBarSeries (weekly conversion funnel).
- **QProcess:** launch morning_brief.py from GUI, real-time stdout in QTextEdit, QSystemTrayIcon notifications.
- **Senior patterns:** Q_LOGGING_CATEGORY, qScopeGuard RAIL, Q_DISABLE_COPY_MOVE, [[nodiscard]], DiagnosticsDialog.
- **Test isolation:** SQLite :memory: via named connection — no production PostgreSQL in CI.

eBPF System Monitor — Linux kernel tracing (kprobe/XDP)

July 2026

Stack: C · C++20 · libbpf · XDP · kprobe/tracepoint · std::jthread ThreadPool · GitHub Actions CI

github.com/stanislaw-perfilyev/ebpf-system-monitor

- TCP connection tracer: kprobe on tcp_connect/tcp_close → kernel ring buffer → async C++ userspace reader (epoll edge-triggered).
- XDP packet counter broken down by L3/L4 protocol; syscall auditor via tracepoint sys_enter (execve, ptrace, process_vm_read/write).
- Userspace: ThreadPools on std::jthread + condition_variable, RequestStats on std::shared_mutex (lock-free-for-readers stats snapshot), RAIL RequestTimer.

- 51 GTests (BPF programs via mocked ring buffer, userspace logic without root) · CI: GitHub Actions.

Linux Kernel Driver — 4 loadable kernel modules (LKM)

July 2026

Stack: C · Linux Kernel Module · ioctl · procfs/seq_file · Netfilter · kthread · C++17 userspace clients
github.com/stanislaw-perfilyev/linux-kernel-driver

- 01_chardev: ring-buffer character device with spinlock, wait_queue for blocking read(), poll(), ioctl (GET_STATS/RESET/SET_FILTER_PID).
- 02_procfs: /proc/mydriver/{stats,config} via the modern proc_ops + seq_file API.
- 03_netfilter: NF_INET_PRE_ROUTING hook — packet filtering by IP/port, atomic64_t accept/drop/error counters.
- 04_kthread: background kernel thread (kthread_run + workqueue) collects memory snapshots via si_meminfo().
- C++17 RAII clients for every module + bash integration tests · CI: GitHub Actions.

gRPC C++ Microservice — 4 RPC patterns + PVS-Studio

July 2026

Stack: C++17 · gRPC · Protocol Buffers · std::shared_mutex · GitLab CI · PVS-Studio
github.com/stanislaw-perfilyev/grpc-vacancy-service

- All 4 gRPC communication patterns implemented: unary, server-streaming, client-streaming, bidirectional-streaming.
- Thread-safe in-memory store: std::shared_mutex (write → unique_lock, snapshot → shared_lock) + atomic counters.
- gRPC interceptor: per-call logging and latency measurement; server health-check and thread pool.
- C++ RAII client and CLI demo, plus a Python client covering every RPC type.
- 20 GTests (6 multithreading tests) · CI: GitLab CI + PVS-Studio static analysis.

Robot Teacher — ROS2 (Computer Vision + Speech + LLM)

July 2026

Stack: C++17 · ROS2 Jazzy · OpenCV · Groq Whisper API · LLM (Llama 3.1) · ament_lint · GTest
github.com/stanislaw-perfilyev/Robot

- 3 independent C++ ROS2 nodes: perception_node (V4L2 capture, Haar face detection, OpenCV-contour classification of 8 hand gestures), hearing_node (voice capture + Groq Whisper STT), dialog_node (6-state finite state machine + LLM dialogue via the Groq API).
- Web UI (SSE + MJPEG stream) renders the camera feed, dialogue, and session state in real time.
- Full ament_lint compliance: cpplint, pep257, uncrustify, copyright headers; 22 GTests (gesture classifier, dialogue logic, WAV utilities).
- GitHub Actions CI (colcon build + colcon test) · tested on Ubuntu 24.04 + ROS2 Jazzy.

DAQ Bug Fix — code review (cpp-code-review)

June 2026

Stack: C++ · Qt · QThread/QMutex/QWaitCondition · Code Review
github.com/stanislaw-perfilyev/cpp-code-review

- Found and fixed a double-free in cleanup() (pointer wasn't nulled after delete[]) and a memory leak on partial array allocation during std::bad_alloc in the USB DAQ write threads; producer-consumer pattern on QMutex/QWaitCondition.

PORTFOLIO — PYTHON / BACKEND

Job Search Automation System — Production Backend

April 2026 — present

Stack: FastAPI · PostgreSQL · Redis · asyncio · Docker · Railway · WebSocket · GitHub Actions
github.com/stanislaw-perfilyev/job-search-automation-portfolio

- Vacancy aggregation: hh.kz REST API + Habr Career RSS + 8 Telegram channels (~100 vacancies/day), deduplication, morning brief at 09:00 Almaty.
- FastAPI production: REST + WebSocket (/ws/updates) + Redis TTL cache + APScheduler cron + logging middleware. Deployed on Railway.
- PostgreSQL schema: vacancies + freelance + kpi_events; SQLAlchemy ORM + asyncpg; ClickHouse-ready OLAP design.
- LLM-powered cover letter generator (Claude API): A/B/C templates, --top flag for senior-level premium letters.
- 4 GitHub Actions cron bots: Upwork IMAP monitor, weekly KPI report to Telegram, PG → Sheets sync.
- 100+ pytest tests: monkeypatch, tmp_path, mock for all external APIs.

Custom CI Quality Gate — grand_check.py

2026

Stack: Python · ASAN/TSAN · PVS-Studio · cppcheck · gcov · git hooks

- Custom quality-gate tool: 166+ automated checks — sanitizers (ASAN/TSAN), static analysis (PVS-Studio, cppcheck, cpplint/ament_lint), MT anti-pattern detection (raw lock without RAII, atomic without memory_order), test coverage (gcov).
- Runs on 20+ repositories before every commit; git pre-commit/post-commit hooks trigger it automatically and log metrics to PostgreSQL.

- Result: 7 C++ repositories reached FAIL:0 WARN:0 ("ideal") status; 450+ automated tests (GTest) pass across the portfolio.

PORTFOLIO — SYSTEMS / MIXED

WinAPI Showcase · QML System Monitor · D-Bus Service

June 2026

Stack: WinAPI · DLL injection · QML/Qt Quick · D-Bus · Linux sysfs · POSIX API

github.com/stanislav-perfilyev/job-search-automation-portfolio/tree/main/portfolio

- **WinAPI Showcase:** 3 DLP programs — Clipboard monitoring (SetWindowsHookEx), DLL injection (CreateRemoteThread), filesystem monitoring (ReadDirectoryChangesW). Concept comparable to ProfiStaff DLP (400K+ users).
- **QML System Monitor:** Qt Quick + QML, real-time CPU/RAM/Disk metrics via /proc, animated SVG gauges, POSIX API (sysinfo, getrusage).
- **D-Bus Service:** org.perfilyev.SystemMonitor interface, GetCpuLoad / GetMemoryInfo / StartMonitor via sd-bus C API.

MIL-STD-1553B Protocol Analyzer — Embedded / Defence

June 2026

Stack: Object Pascal (Delphi) · MIL-STD-1553B · DUnit · State Machine

github.com/stanislav-perfilyev/job-search-automation-portfolio/tree/main/portfolio/mil1553_analyzer

- Bit-accurate decoder + simulator of MIL-STD-1553B: BC/RT/BM modes, full state machine, error injection.
- 40+ DUnit tests. Applicable to aerospace, defence, and avionics embedded systems.

WORK EXPERIENCE

CNC Software Engineer / Embedded Developer

July 2015 — November 2025

Self-employed / Freelance · Almaty, Kazakhstan

- C++/Qt: control software for CNC systems — G-code interpreters, axis control algorithms, EDR data logger (QThread + producer-consumer on QMutex/QWaitCondition).
- Real-time USB sensor data collection: X/Y/Z axes, temperature, battery, humidity. Circular event buffer, async UDF file writing, thread-safe graceful shutdown.
- Software simulation of equipment: axis motion models and algorithm verification before hardware deployment.
- Full project lifecycle: requirements → implementation → delivery → support. 10 years of independent work without supervision.

Tools: Qt 5, QThread, USB communication, SolidWorks, PowerMill, G-code, Siemens/Fanuc controllers

EDUCATION

C++ Developer Course (18 months)

November 2025 — present

Skillbox · C++11/14/17/20, Qt, ROS2, algorithms, templates, Google Test

Bachelor's Degree — Management

University of International Business (UIB), Almaty